

Probability & Statistics Infographics Study Deck

15 enriched infographic slides compiled into one presentation

Module 1: Foundations of Statistics

- Introduction to Machine Learning with Statistics
- Nature of Statistics: Collect, Organize, Summarize, Analyze, Conclude
- Statistics Foundation: Variables, Data, Dataset, and Random Variables
- Descriptive vs Inferential Statistics
- Population vs Sample in Statistical Studies

Module 2: Types of Data and Measurement

- Types of Data: Qualitative and Quantitative Variables
- Discrete vs Continuous Data
- Four Levels of Measurement: Nominal, Ordinal, Interval, Ratio

Module 3: Data Collection and Sampling

- Data Collection Methods in Statistics
- Sampling Techniques: Random, Systematic, Stratified, Cluster, Convenience
- Common Sampling Errors and Biases
- Sampling Technique Practice: Identify the Sampling Method

Module 4: Study Design and Application

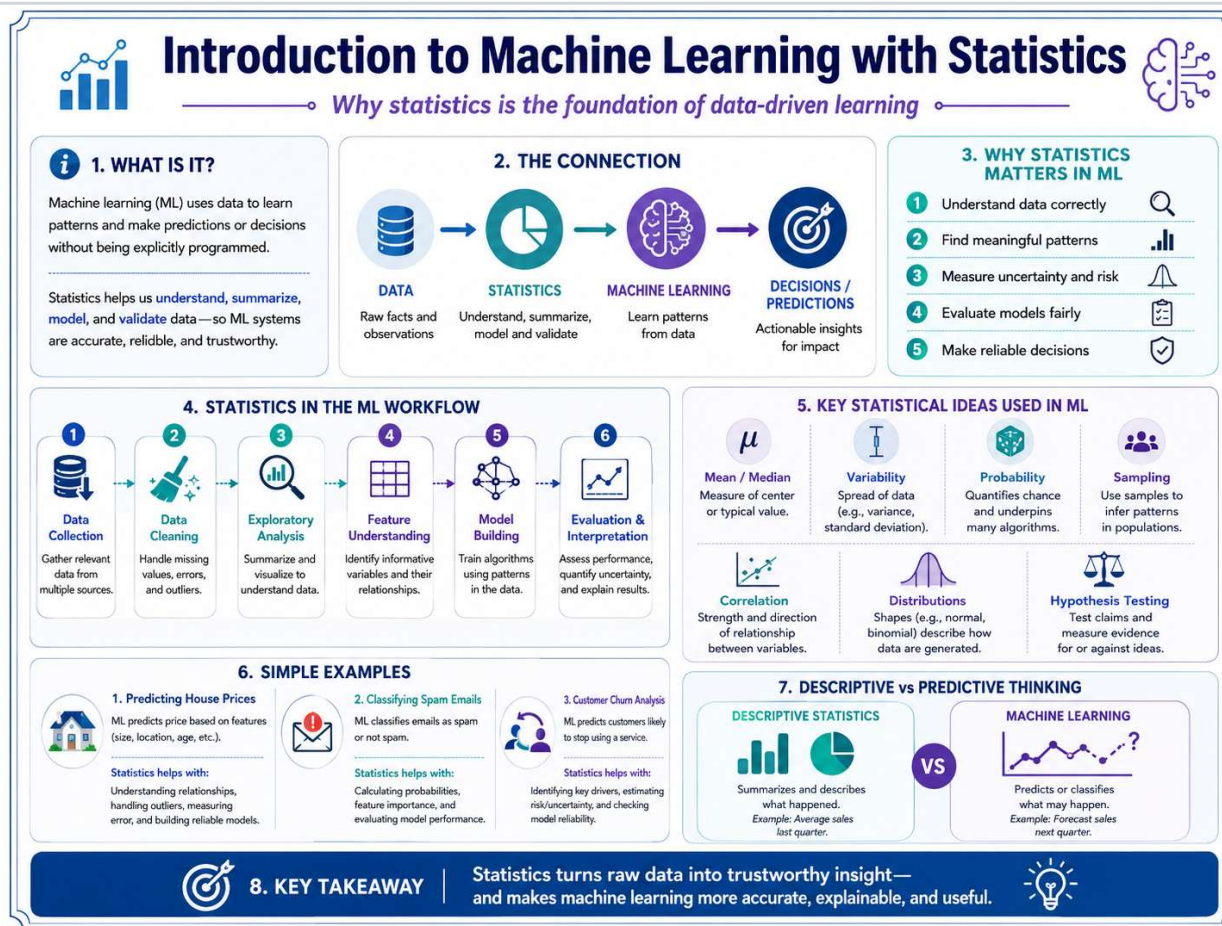
- Observational vs Experimental Studies
- Statistics to Machine Learning Bridge
- Choosing the Right Statistical Method: A Beginner Decision Map

Module 1: Foundations of Statistics

5 infographic slides

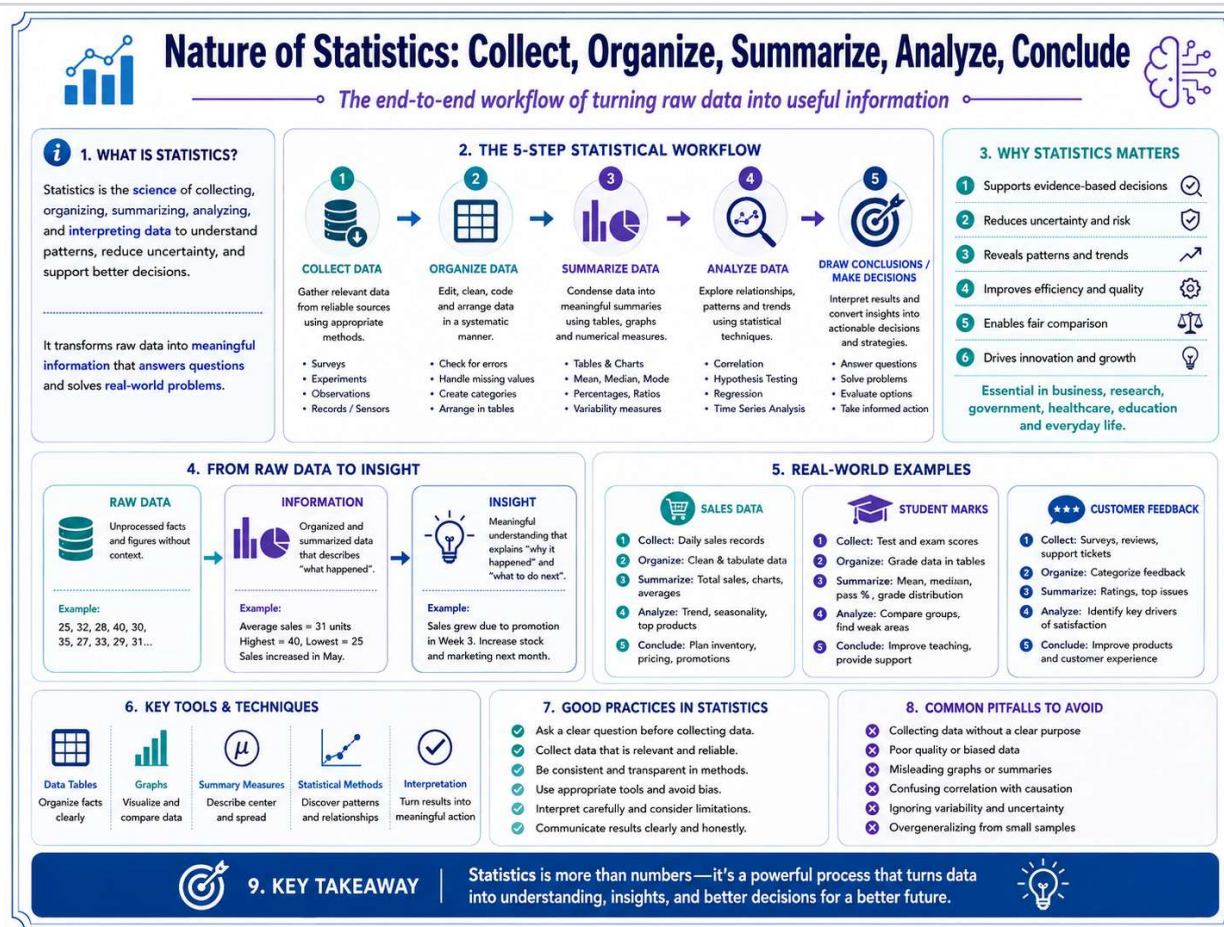
Introduction to Machine Learning with Statistics

Module 1: Foundations of Statistics



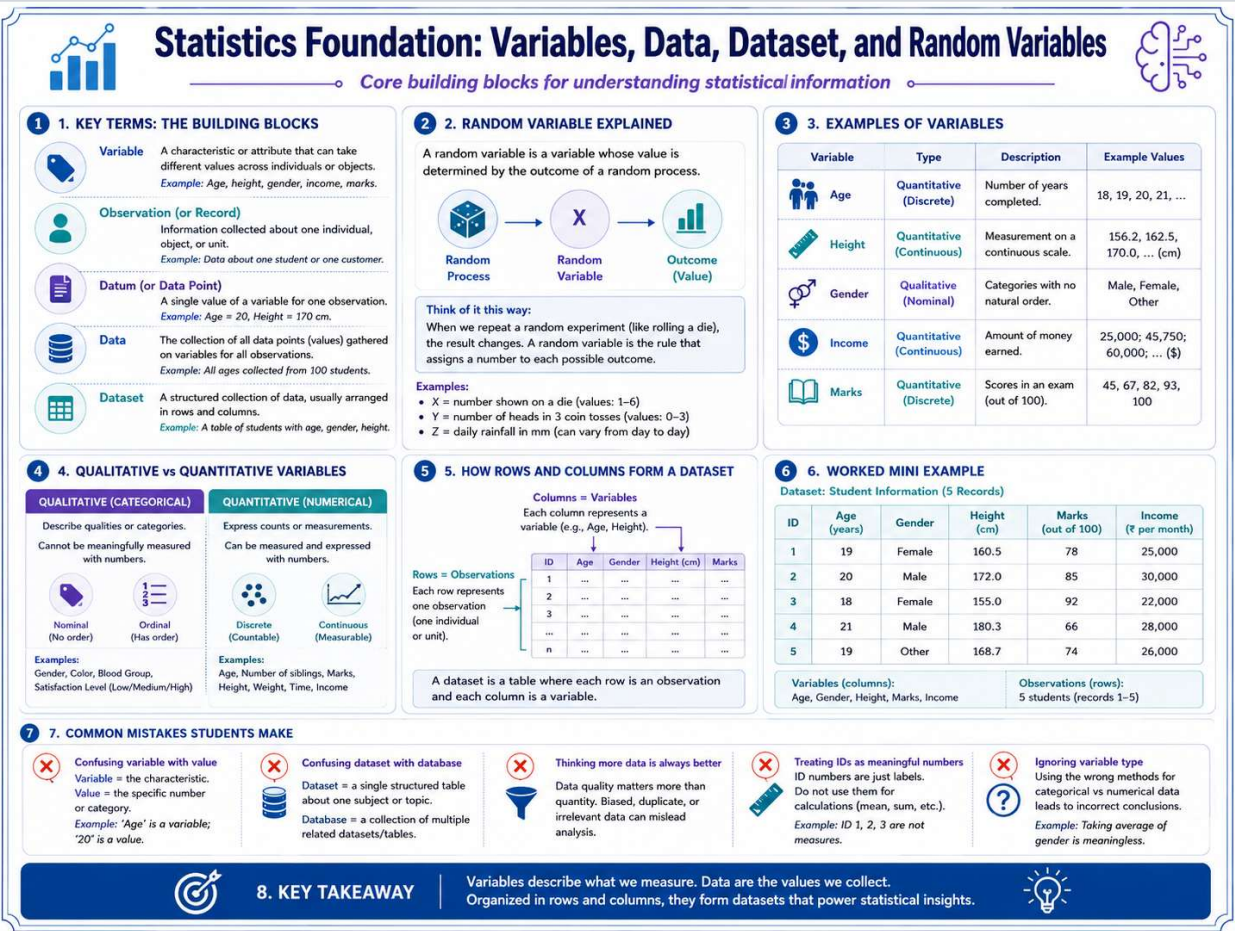
Nature of Statistics: Collect, Organize, Summarize, Analyze, Conclude

Module 1: Foundations of Statistics



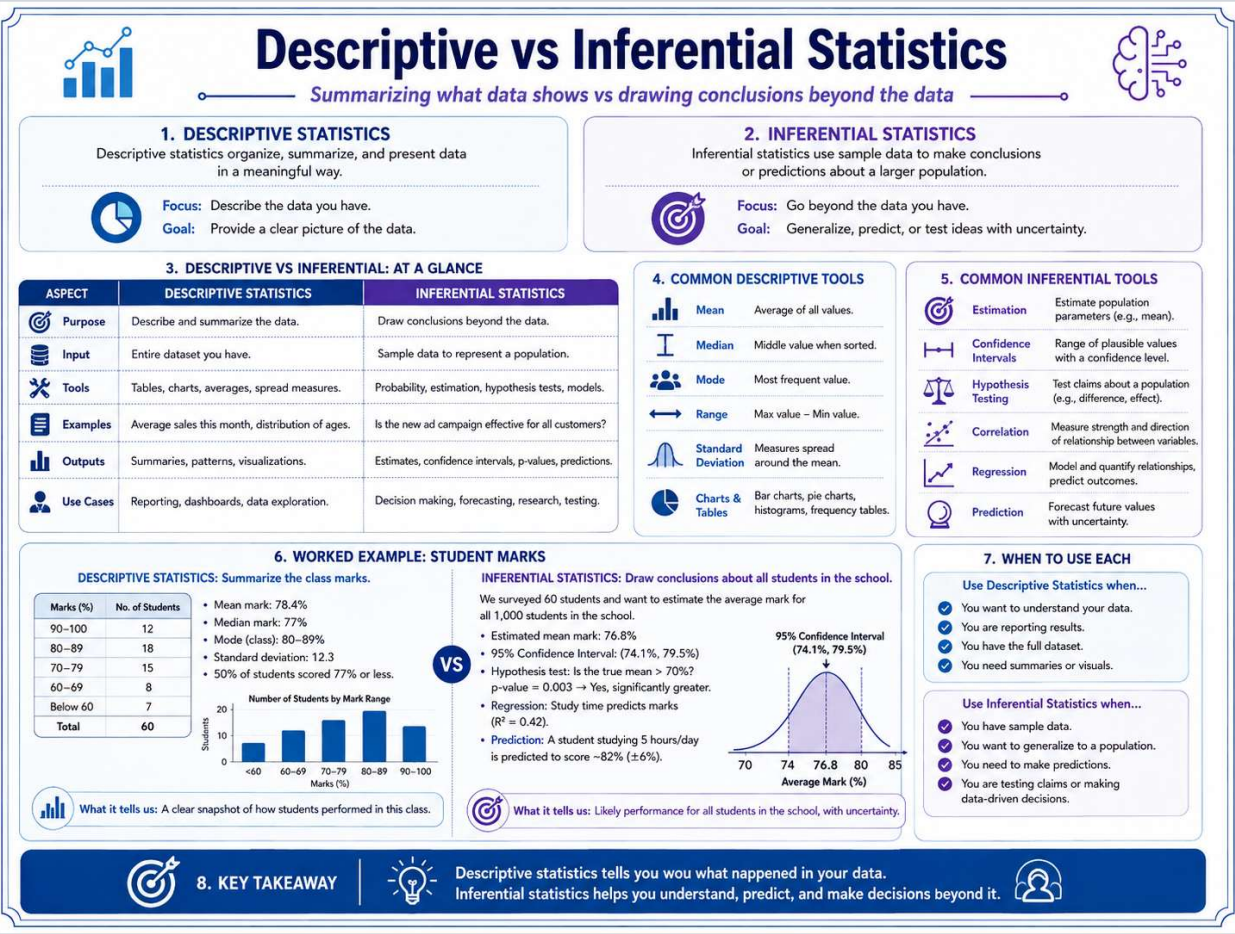
Statistics Foundation: Variables, Data, Dataset, and Random Variables

Module 1: Foundations of Statistics



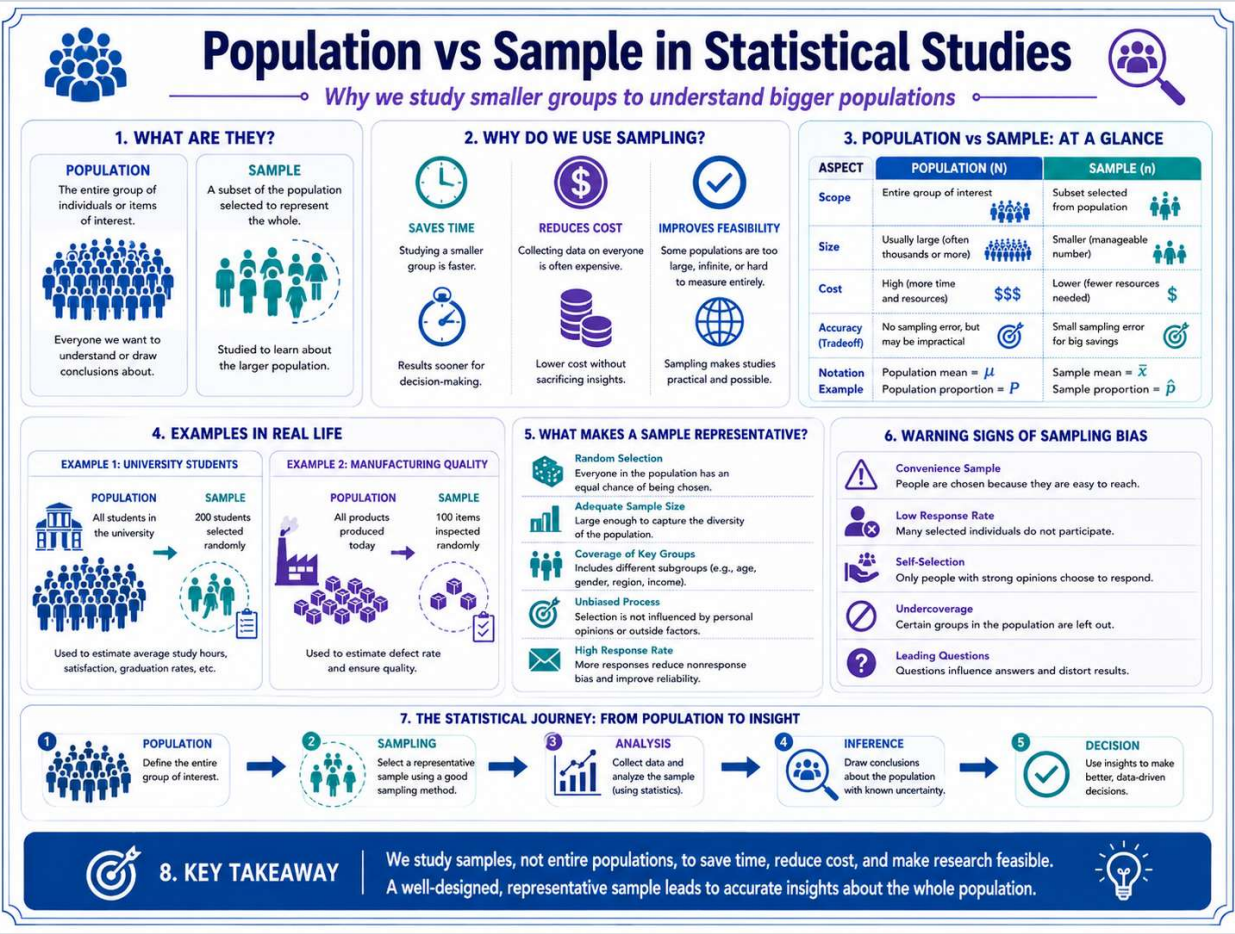
Descriptive vs Inferential Statistics

Module 1: Foundations of Statistics



Population vs Sample in Statistical Studies

Module 1: Foundations of Statistics

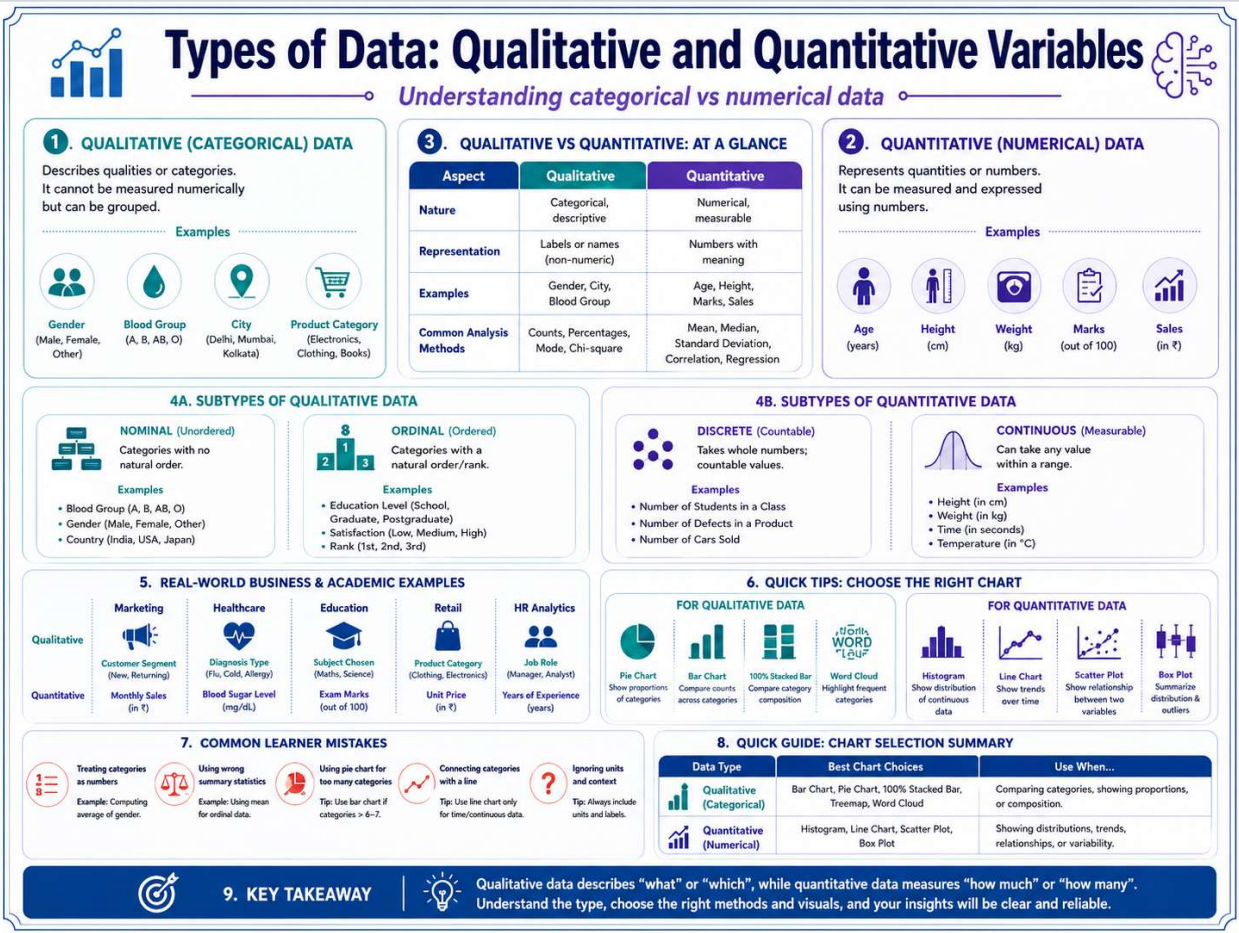


Module 2: Types of Data and Measurement

3 infographic slides

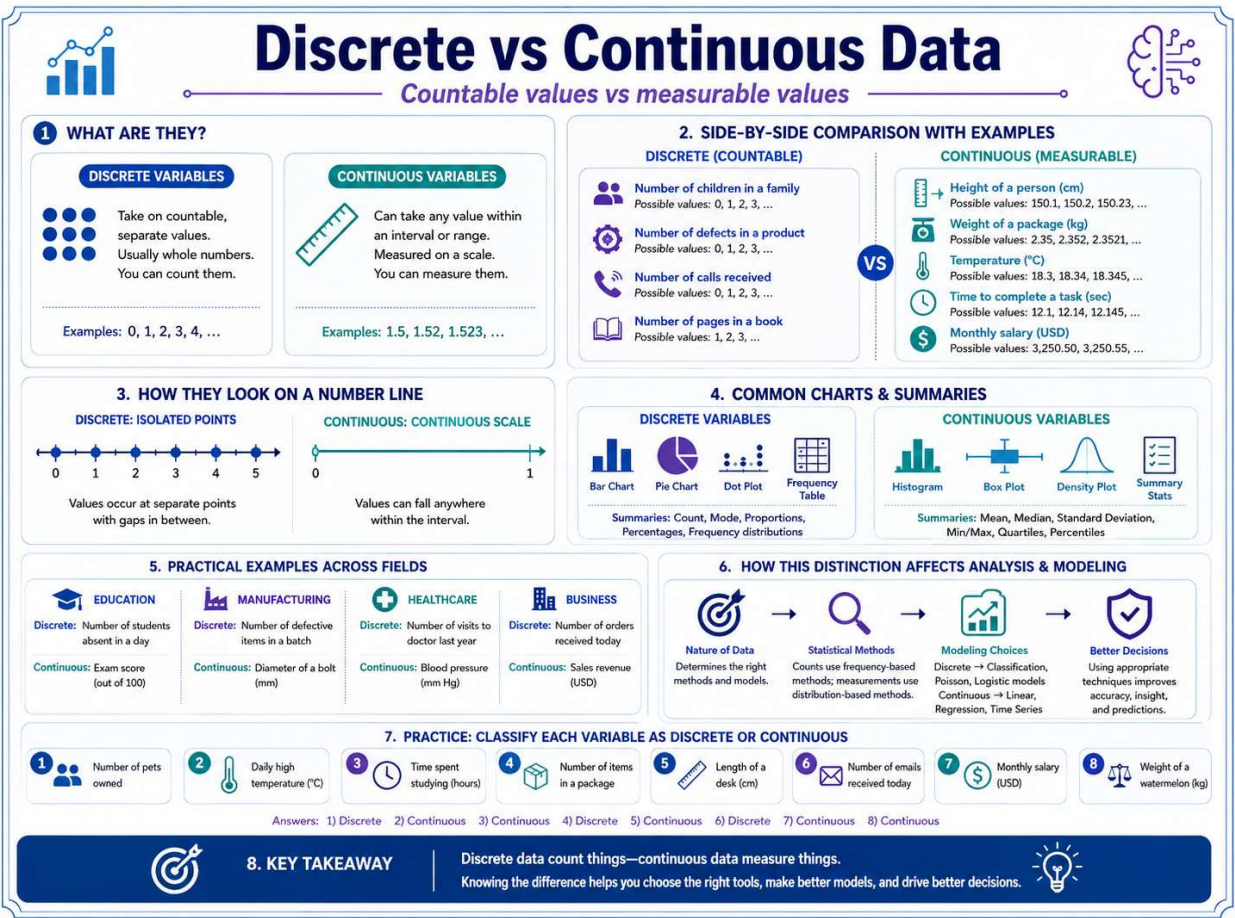
Types of Data: Qualitative and Quantitative Variables

Module 2: Types of Data and Measurement



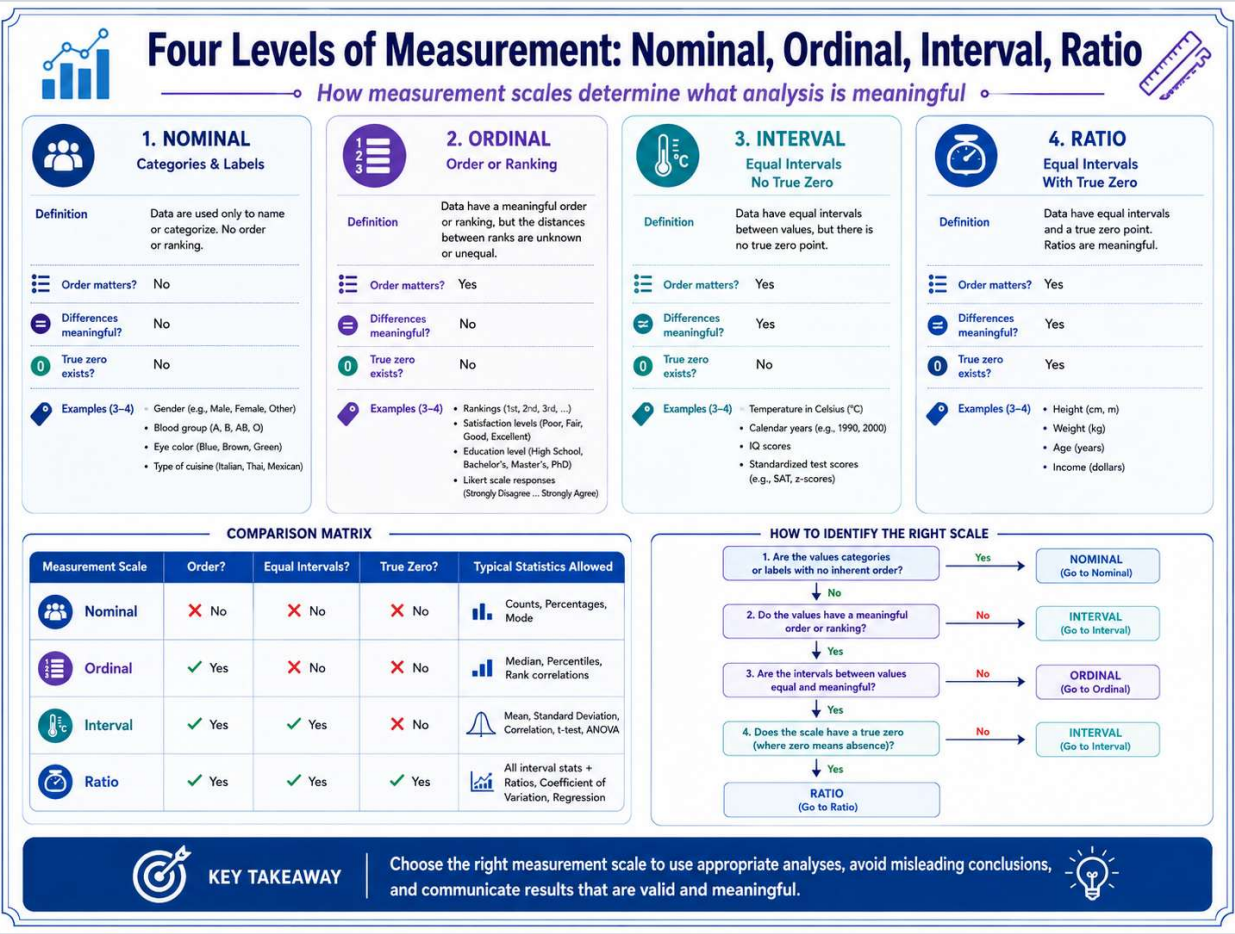
Discrete vs Continuous Data

Module 2: Types of Data and Measurement



Four Levels of Measurement: Nominal, Ordinal, Interval, Ratio

Module 2: Types of Data and Measurement

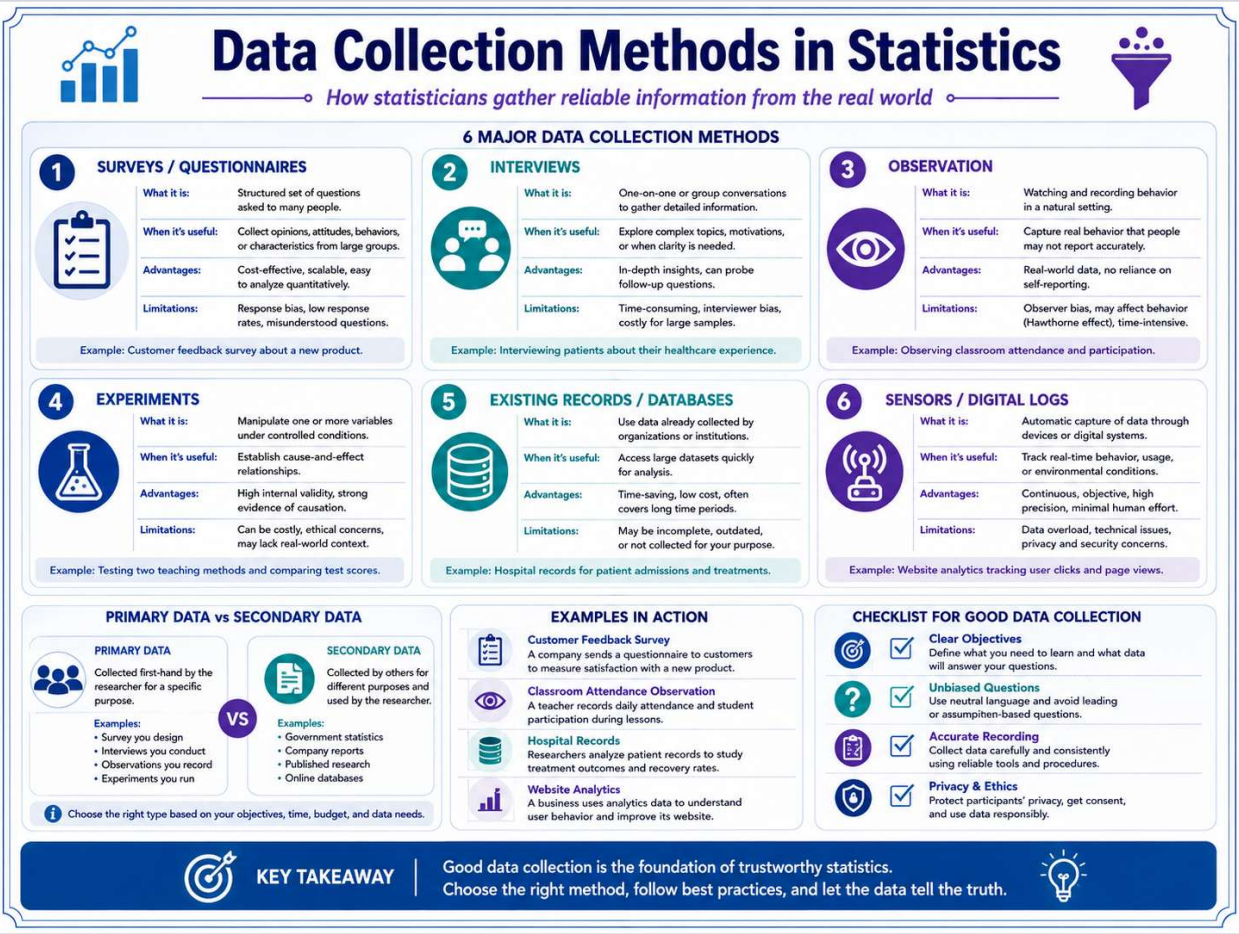


Module 3: Data Collection and Sampling

4 infographic slides

Data Collection Methods in Statistics

Module 3: Data Collection and Sampling



Sampling Techniques: Random, Systematic, Stratified, Cluster, Convenience

Module 3: Data Collection and Sampling

Sampling Techniques: Random, Systematic, Stratified, Cluster, Convenience

Common ways to select a sample from a population

1. WHAT IS SAMPLING?

Sampling is the process of selecting a subset of individuals from a population to study and draw conclusions about the whole population.

WHY WE SAMPLE

Saves time

Lowers cost

Feasible for large populations

Can be more accurate with careful design

Enables faster decisions

GOOD SAMPLING = BETTER INSIGHTS

The quality of your sample affects the accuracy, reliability, and usefulness of your results.

2. SIMPLE RANDOM SAMPLING

Every member has an equal chance of being selected. Use random numbers or a randomizer to choose.

When the population is homogeneous and a complete list of all members is available.

A school randomly selects 50 students from an enrollment list of 2,000 to survey study habits.

May be costly or impractical for very large or widely dispersed populations.

3. SYSTEMATIC SAMPLING

Pick a random starting point and then select every k-th member from an ordered list.

When the population has a natural order or sequence.

A factory inspects every 10th item on the production line starting with the 4th item.

Can be biased if there is a hidden pattern in the order that matches the interval (k).

4. STRATIFIED SAMPLING

Divide the population into meaningful subgroups (strata) then randomly sample from each stratum.

When subgroups differ and you want each represented in the sample.

A company surveys employees by department to ensure each department is proportionally represented.

Requires information to define strata; more planning and data needed.

5. CLUSTER SAMPLING

Divide the population into clusters (natural groups), randomly select clusters, then survey all or some within them.

When populations are spread out and a full list of individuals is hard to obtain.

A health study selects random neighborhoods, then surveys all households in those neighborhoods.

Usually less precise than random sampling due to similarity within clusters.

6. CONVENIENCE SAMPLING

Choose members who are easily accessible and willing to participate.

When time, budget, or access is very limited; for pilot studies or exploratory work.

A researcher surveys customers who are available at a mall kiosk.

High risk of bias; results may not represent the broader population.

7. COMPARISON AT A GLANCE

Sampling Method	Fairness (Chance for all)	Cost	Complexity	Representativeness	Best Use
2 Simple Random	★★★★★	\$\$	Low	High	When full list exists and population is fairly uniform.
3 Systematic	★★★★☆	\$	Low	Medium-High*	When ordered list exists and no hidden pattern.
4 Stratified	★★★★☆	\$\$	Medium	High	When subgroups differ and must be represented.
5 Cluster	★★★☆☆	\$	Medium	Medium	When population is geographically spread out.
6 Convenience	★☆☆☆☆	\$	Very Low	Low	For quick, low-cost, exploratory studies.

*Systematic sampling can be highly representative if the interval does not align with a pattern.

8. CHOOSE THE RIGHT SAMPLING METHOD

Do you have a complete list of all members?> 2 Simple Random

Is the population in a natural order/sequence?> 3 Systematic

Are there distinct subgroups you want represented?> 4 Stratified

Is the population large and spread out in natural groups or locations?> 5 Cluster

Are time, cost, or access very limited?> 6 Convenience

9. KEY TAKEAWAY

The right sampling technique balances accuracy, cost, time, and practicality—leading to trustworthy results and better decisions.

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Common Sampling Errors and Biases

Module 3: Data Collection and Sampling

**STATISTICS
STUDY SERIES**

Common Sampling Errors and Biases

Good data begins with good samples. *Know the risks. Improve the results.*





1 WHAT IS SAMPLING ERROR?

Sampling error is the natural difference that occurs between a sample result and the true population value because we observe a sample instead of the entire population. It is due to chance.

It can be measured and reduced by using better sampling methods and larger sample sizes.



2 WHAT IS BIAS?

Bias is a systematic error that causes a sample result to be consistently different from the true population value in one direction. It is not due to chance and will not improve by increasing sample size.

Bias threatens validity and leads to misleading or unfair conclusions.

3 COMMON SAMPLING ERRORS AND BIASES



1. SAMPLING ERROR (Random Error)

Chance differences between sample results and the true population value.

Can be reduced with larger, better-designed random samples.



2. SELECTION BIAS

Systematic differences in how individuals are chosen for the sample.

Leads to samples that are not representative of the population.



3. CONVENIENCE BIAS

Selecting participants who are easiest or most convenient to reach.

Often results in unrepresentative samples.



4. NONRESPONSE BIAS

Occurs when people who do not respond are different from those who do.

Can distort results if nonrespondents have different views.



5. UNDERCOVERAGE (BIAS)

Some members of the population have little or no chance of being included in the sample.

Examples: no internet access, missing phone numbers, etc.



6. RESPONSE BIAS

Respondents give answers that are untruthful, inaccurate, or influenced by how the question is asked.

Includes social desirability bias.



7. INTERVIEWER BIAS

Interviewer's tone, behavior, or reactions influence how respondents answer.

Can affect responses consciously or unconsciously.



8. SURVIVORSHIP BIAS

Focusing only on those who "survived" or remained, ignoring those who did not.

Leads to overly optimistic conclusions.

4 RANDOM ERROR vs SYSTEMATIC BIAS

RANDOM ERROR (Good News!)	SYSTEMATIC BIAS (Bad News!)
Caused by chance variation in sampling.	Caused by flaws in the way the sample is selected, collected, or respondents behave.
Results vary around the true value (sometimes high, sometimes low).	Results consistently off in one direction (too high or too low).
Decreases as sample size increases.	Does NOT decrease with larger sample size.
Affects precision.	Affects accuracy (validity).
Example: Flipping a coin 100 times may not be exactly 50/50.	Example: Only surveying people who like a product will overstate how popular it is.

5 REAL-WORLD EXAMPLES

Online Survey	Mail-Intercept Survey	Political Polling	Product Feedback	Classroom Sampling
 People without internet access are excluded (undercoverage). Those with strong opinions may be more likely to respond (nonresponse bias).	 Only people at the mall are included (convenience bias). Not representative of the general population.	 Likely voters only may exclude disengaged citizens (undercoverage). Question wording can create response bias.	 Mostly satisfied customers respond; dissatisfied ones stay silent (nonresponse bias). Overstates satisfaction.	 Surveying one easy class or study group causes selection bias. Results may not apply to the entire student body.

6 HOW TO REDUCE BIAS AND ERROR



1. Proper Sample Size

Use a sample large enough to reduce random error and increase precision.



2. Random Selection

Give every member of the population an equal chance of being selected.



3. Stratification

Divide the population into relevant subgroups and sample from each proportionally.



4. Neutral Wording

Use clear, unbiased language. Avoid leading or loaded questions.



5. Follow-Ups

Encourage and remind nonrespondents to reduce nonresponse bias.



6. Pilot Testing

Test questions and procedures on a small group before the full study.

7 QUICK WARNING SIGNS CHECKLIST

- ✓ Sample is small.
- ✓ Sample is from only one place or group.
- ✓ Participation is voluntary without follow-up.
- ✓ Large differences between respondents and nonrespondents.
- ✓ Questions are leading, double-barreled, or confusing.
- ✓ Interviewer behavior may influence answers.
- ✓ Results seem too good to be true or only show one side.

**KEY TAKEAWAY:** Sampling error is about chance; bias is about process.

| Use sound methods to get accurate, reliable, and trustworthy results.



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Sampling Technique Practice: Identify the Sampling Method

Module 3: Data Collection and Sampling

Sampling Technique Practice: Identify the Sampling Method

PRACTICE • RECOGNIZE • REVISE • SUCCEED

1 RECAP: THE FIVE SAMPLING METHODS

1 Simple Random Sampling

Everyone has an equal chance of being selected.

2 Systematic Sampling

Select every k^{th} member after a random start.

3 Stratified Sampling

Divide into groups (strata) and sample from each group.

4 Cluster Sampling

Randomly select clusters, then survey all in those clusters.

5 Convenience Sampling

Choose whoever is easily available or convenient.

2 DECISION CLUES: HOW TO RECOGNIZE EACH METHOD

1 **Simple Random:** Based on chance; no pattern; equal opportunity.

2 **Systematic:** Look for "every k^{th} " or a fixed interval.

3 **Stratified:** Population is divided into meaningful subgroups (strata).

4 **Cluster:** Groups (clusters) are selected; often entire clusters are used.

5 **Convenience:** Selected due to ease of access or availability.

3 PRACTICE: IDENTIFY THE SAMPLING METHOD

Read each scenario and identify the sampling method used.

1 A school has a list of all 600 students. Using a random number generator, 50 students are selected.

Answer: Simple Random Sampling

2 At a store, the first customer is chosen randomly. After that, every 10th customer is surveyed.

Answer: Systematic Sampling

3 Employees are divided by department (HR, Sales, Finance). A random sample is taken from each.

Answer: Stratified Sampling

4 Two neighborhoods are randomly chosen. Every household in those neighborhoods is surveyed.

Answer: Cluster Sampling

5 A survey is conducted by asking whoever is nearby in the mall to participate.

Answer: Convenience Sampling

6 From a jar containing employee ID slips, 75 slips are mixed and 25 are drawn at random.

Answer: Simple Random Sampling

7 Starting with the 5th name on a list, every 15th name is selected until the sample size is met.

Answer: Systematic Sampling

8 The city is divided into age groups (18–25, 26–40, 41–60, 60+). A sample is taken from each group.

Answer: Stratified Sampling

9 Three schools are randomly selected from the district. All students in those schools are surveyed.

Answer: Cluster Sampling

10 Patients are surveyed as they leave a clinic because they are easy to find and willing to respond.

Answer: Convenience Sampling

11 A list of 1,000 households is used. 200 households are selected randomly using a computer.

Answer: Simple Random Sampling

12 Every 10th item is picked from an inventory list after a random start.

Answer: Systematic Sampling

4 MINI CHALLENGE: WHAT'S THE METHOD?

Identify the sampling method for each scenario.

A A company divides its branches by region. From each region, 5 branches are randomly selected. All customers in those branches are surveyed.

Answer: _____

B In a factory, starting with the 7th product off the line, every 25th product is inspected.

Answer: _____

C Students who are waiting outside the library are asked to fill out a short feedback form.

Answer: _____

COMMON MISTAKES TO AVOID

- Confusing systematic with stratified – Systematic uses a fixed interval, stratified uses groups (e.g., by age, department).
- Confusing cluster with stratified – Cluster selects whole groups (e.g., schools, neighborhoods), stratified samples within each group.
- Assuming convenience is random – Convenience is based on ease, not chance.
- Overlooking the random start in systematic sampling.
- Forgetting that in cluster sampling, all (or many) elements in the selected clusters are included.

ANSWER KEY

Practice (3):

1. Simple Random	2. Systematic	3. Stratified
4. Cluster	5. Convenience	6. Simple Random
7. Systematic	8. Stratified	9. Cluster
10. Convenience	11. Simple Random	12. Systematic

Mini Challenge (4):

A. Cluster	B. Systematic	C. Convenience
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(Check your reasoning using the decision clues above!)

KEY TAKEAWAY: Look for patterns, groups, clustering, or ease of access. Use the decision clues to match each scenario to the correct sampling method!

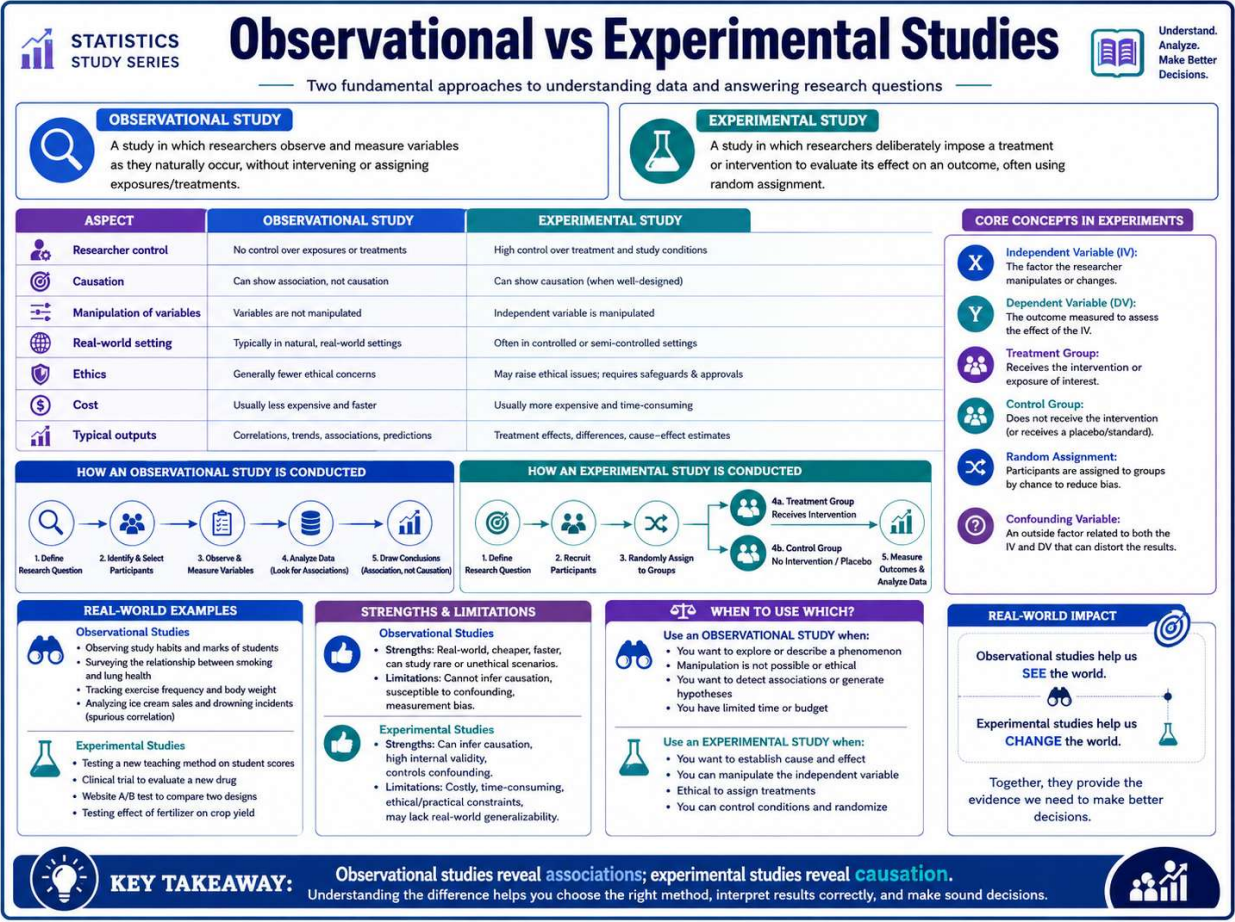
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Module 4: Study Design and Application

3 infographic slides

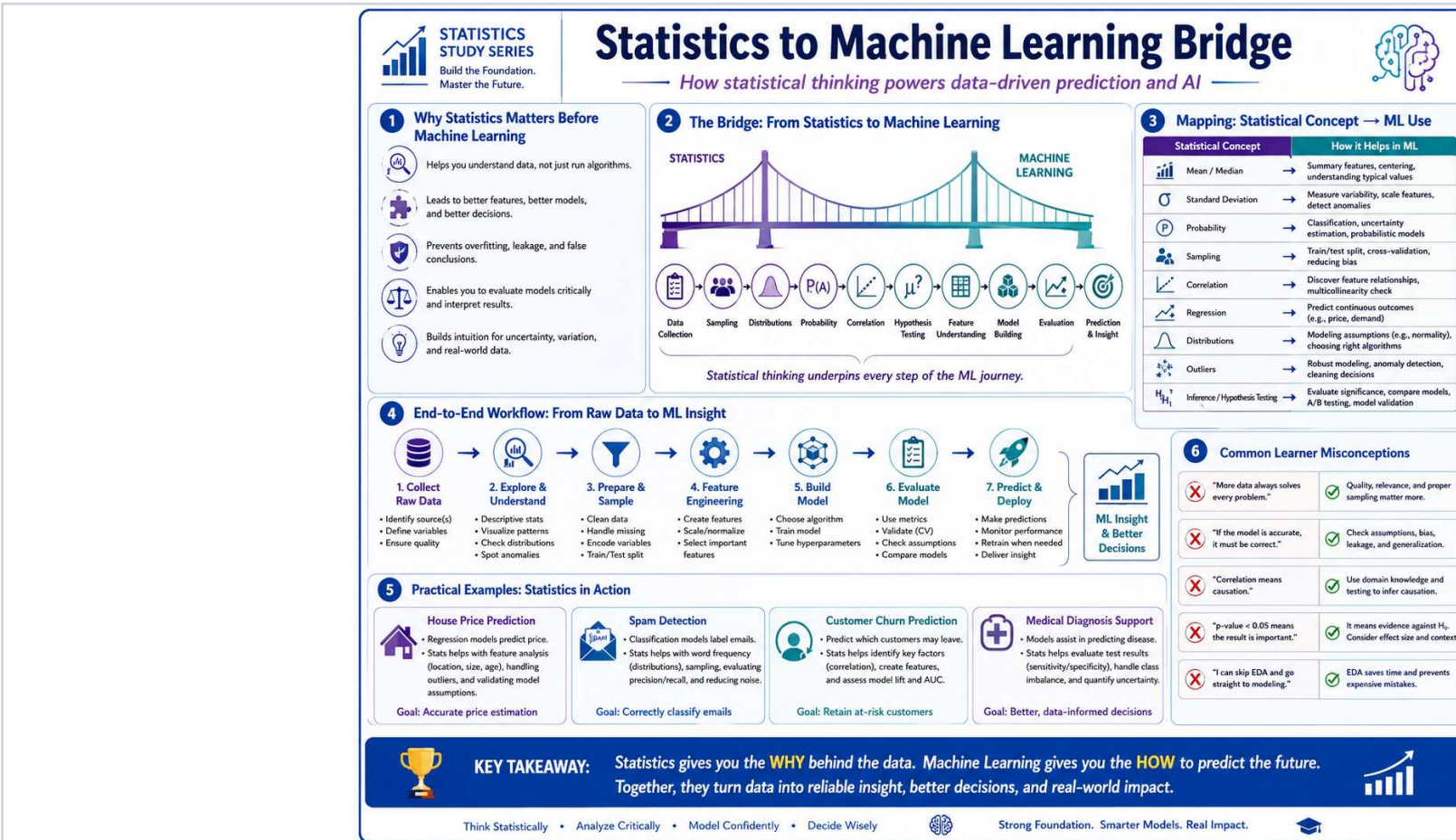
Observational vs Experimental Studies

Module 4: Study Design and Application



Statistics to Machine Learning Bridge

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